2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.203 Computer Networks

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

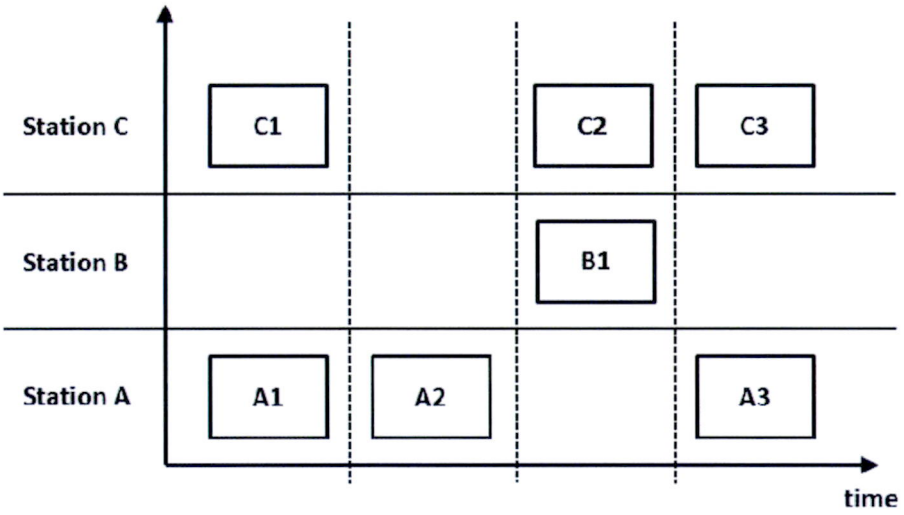
1.a Communication is organised in multiple layers. Give two reasons why communication systems are organised in layers. Further, outline for each of the four lower layers (in the correct order of the ISO/OSI reference model) what their function is and explain how they are used by the layer above.

[**6 marks**]

1.b For digital transmission on the Physical Layer different coding techniques can be used. Given the following bit stream: 10100011. Sketch the encoding of this bit stream according to Manchester and Differential Manchester encoding. What are the main advantages and disadvantages of these coding techniques?

[**6 marks**]

1.c Multiplexing is used to transmit multiple conversations over the same cable. With Time Division Multiplexing (TDM) timeslots are allocated to specific conversations. In the following figure, three stations A, B, and C want to send data. Show the multiplexed data when a) Synchronous b) Asynchronous multiplexing is used. Do not forget to insert headers and empty slots where they are necessary. Note, if packets arrive at the same time, process them in alphabetical order.



[**4 marks**]

Question 1 continues on next page...

Question 1 continued…

1.d Medium Access Control at the Data Link Layer is required to control when a computer transmits a message.

i. Compare TDMA (Time Division Multiple Access) with CSMA (Carrier Sense Multiple Access) with respect to efficiency and fairness. (4 marks)

ii. Why can collisions occur, even if a MAC protocol with "Carrier Sense" (CSMA) is used? (2 marks)

iii. Collisions especially occur under high load when too many nodes try to send at the same time. Propose a solution that offers a minimum guaranteed level of service to nodes. Also discuss potential disadvantages of your solution. (3 marks)

[**9 marks**]

Total 25 marks

Question 2

2.a For connection-oriented services circuit switching and virtual circuit switching can be used. Describe the two approaches and explain their differences. Which one can be more easily deployed in a packet switched network?

[**4 marks**]

2.b Multiprotocol Label Switching (MPLS) uses fixed length labels instead of IP addresses. Describe how MPLS works (use a diagram if you wish) and discuss advantages and disadvantages of MPLS compared to standard IP routing.

[**5 marks**]

2.c Due to the immense growth of connected devices the Internet is running out of free addresses. Network Address Translation (NAT) is used in order to deal with the shortage of IPv4 addresses.

i. Describe how NAT addresses this issue and discuss advantages and disadvantages of this approach. (3 marks)

ii. IPv6 is also dealing with the address shortage. Outline how this problem would be solve through use of IPv6. (2marks)

iii. You are working for an organisation that runs its own sub-network. So far they have used NAT to deal with the IP address problem. Because of an increasing number of mobile devices within their network and also multi-homing devices they are considering switching to IPv6. They have asked you to discuss the advantages and disadvantages of this switch over. (3 marks)

[**8 marks**]

Question 2 continues on next page…

Question 2 continued...

2.d The *Domain Name Service* (DNS) is located at the Application Layer of the Internet Protocol suite and deals with the translation of Internet Names to IP Addresses.

i. Explain the role of Domain Names within DNS. Specifically explain their structure and what they represent. (2 marks)

ii. You are administering the domain ulanc.co.uk and the marketing department has decided that they need an address called www.marketing@ulancs.co.uk for their department.

Will you be able to create such a domain? If not, what will you propose instead? (2marks)

[**4 marks**]

2.e The Transport layer is the lowest level end-to-end protocol.

i. Describe the role of the transport layer, specifically explain the two types of transport services and name corresponding protocols and applications making use of the different service types (2 marks)

ii. For addressing at the transport layer a combination of IP address and Port number is used. Why cannot simply the process ID of the corresponding processes be used? (2 marks)

[**4 marks**]

Total 25 marks

Question 3

3.a One of the key design decisions in the development of the Internet protocols has been to be strict when sending and tolerant when receiving. What is meant by this and why is such a design decision important in systems such as the Internet?

[**5 marks**]

3.b You come across two machines with the following network configurations

Machine A		**Machine B**	
IP Address:	148.88.130.4	IP Address:	148.88.135.10/17
Subnet Mask:	255.255.128.0		

i. For each of the above configurations, state whether the machine is using *classless* addresses and in each case, how this can be determined. (1 mark)

ii. Outline the two key roles routers play in the Internet. Then, justifying your answer based on the above machine configurations and showing all working, determine whether these devices would be able to communicate without the need for a router? (6 marks)

iii. If machine A needed to send an Internet Protocol (IP) packet to every device on its network, what IP address would it use as a destination for the packet? Your answer must show all working. (2 marks)

iv. Devices running IP will typically include the address 127.0.0.1 as part of their network configuration. What is this address and what is its purpose? (2 marks)

[**11 marks**]

3.c The Internet Protocol (IP) header includes a *time to live* (TTL) field.

i. Outline how this field is used during the transmission of IP packets and explain the problem the inclusion of this field is trying to avoid? (3 marks)

ii. Being careful to identify any additional protocols needed, and outlining the messages sent and received, explain how the *traceroute* utility makes use of the *time to live* field to find the set of network nodes a packet passes through as it makes its way between the source and destination nodes. (6 marks)

[**9 marks**]

Total 25 marks

Question 4

4.a A fragment-able but currently un-fragmented IPv4 packet with a 200-byte payload arrives at an Internet router that has an MTU of 100 bytes on the outgoing interface.

i. How does the router know that the packet hasn't previously been fragmented but can be fragmented if necessary? (2 marks)

ii. Identify the sequence of packets that will be transmitted by the router when it forwards the packet data. Your answer must clearly identify the amount of payload data and any header values related to the fragmentation process contained within each packet in the sequence. (10 marks)

[**12 marks**]

4.b Routing protocols can be classified as either *interior routing protocols* or *exterior routing protocols*.

i. Identify the main exterior routing protocol used in the Internet. (1 mark)

ii. Outline where such an exterior routing protocol would be used and, giving specific examples to justify your answer, highlight at least three features or constraints exterior protocols support that make them more suitable than interior routing protocols for this purpose.
(7 marks)

[**8 marks**]

4.c What is meant by an *Autonomous System* (AS) within the context of routing protocols and how are *Autonomous System Numbers* used by the routing protocol you identified in 4b part i?

[**5 marks**]

Total 25 marks

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